When Does Spectral Adaptation Help? Information Budgets and Mismatch Costs for Continuous Signal Representations

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Abstract

Spectral adaptation changes which frequencies a continuous representation prefers, yet its benefits vary across reconstruction tasks. We examine this variation through a classical linear-estimation reference conditioned on an observation operator, noise model and assumed signal covariance. Exact scalar and matrix mismatch identities are distinguished from approximate band diagnostics and empirical observations about trained networks. An independent reimplementation supports mismatch-cost ranking (Spearman 0.903 over 60 aggregated cells). A multi-frame diagnostic ranks eight practical GP priors across three noise levels (0.856), but selects the best prior at only one level. Completed seed2/3 evaluations of LTE and SC-INR support simple footprint ordering of readout losses in all four architecture–seed groups. Scale marginalisation predicts ordering in four groups and magnitude in three, but fails function similarity in all four and does not consistently beat the simpler predictor. A prospective 36-case linear-GP test improves a fixed default by 1.105 dB (95% image-bootstrap interval [.726, 1.504]), but chooses the same prior in every case, trails four-fold CV by .112 dB and costs 1.76 times as much. The results support a conditional account of mismatch costs and empirical readout-loss structure, rather than a universal recovery boundary, a proven network mechanism, or a cheap adaptive model selector.

1 Introduction

Continuous (implicit) representations decode a signal at arbitrary coordinates, and a large body of work controls *which frequencies* such a representation can express: the frequency scale of sinusoidal activations [Sitzmann et al., 2020, Liu et al., 2024], the bandwidth of Fourier features [Tancik et al., 2020], target-spectrum-driven hyperparameter selection [Kania et al., 2024, Chandravamsi et al., 2025, Dądela et al., 2026], per-band capacity allocation [Ni et al., 2026], or explicit spectral priors such as Gaussian processes and kriging [Rasmussen and Williams, 2006]. We refer to all of these as *spectral adaptation*. Its empirical record is uneven. In multi-frame super-resolution, choosing the Fourier-feature bandwidth well is decisive [Jyhne et al., 2025]; in sparse-pixel reconstruction, a matched spectrum is worth several decibels (§5.3); but in single-image arbitrary-scale super-resolution (ASISR) the same lever barely moves.

ASISR motivates the question. LIIF and LTE decode local features at continuous coordinates [Chen et al., 2021, Lee and Jin, 2022]; our existing SC-INR research implementation uses an analytic footprint readout. Recent controlled benchmarking reports small gains from more elaborate decoders, alongside substantial dependence on training recipes [Nasir et al., 2026]. Our band measurements show similarity among a limited collection of trained decoders. This is evidence about those models and settings, not a proof of architectural saturation.

We use a classical LMMSE reference conditioned on A, noise and an assumed covariance. A generic $1/f^2$ spectrum permits computation before training, but does not bound all estimators on arbitrary real images. Combined with exact scalar mismatch costs, it supplies conditional tendencies whose band-level accuracy and practical selection value must be tested.

Contributions.

1. **A training-free diagnostic.** We cast spectral adaptation as the choice of a prior spectrum (Prop. 1) and use the per-band linear recoverability R_b^{lin} (Prop. 2) to draw an a-priori three-regime map for single-image, sparse-pixel, multi-frame and 1D signal tasks (§4, Table 1).

2. **A closed-form mismatch cost.** Prop. 3 gives the excess error of a prior that is off by a factor κ in a decoupled scalar channel with recoverability R. Its asymmetry yields a per-band tendency—*wide* is safe as $R \to 1$, *matching* matters at intermediate R, *conservative* is safe where R is low—and its approximate energy-weighted sum over bands empirically ranks GP priors (though it does not reliably pick the single best); it suggests a qualitative interpretation of noisy grid and Fourier-feature fits, without establishing their mechanism (§4).

3. **Pre-registered cross-task validation** (§5). The mismatch-cost shape and asymmetry hold on 60 task–band–κ cells (Spearman 0.903 in the independent reimplementation) [src: E2]; the zones move with the ASISR scale factor as predicted (Spearman 0.952 over 27 cells) [src: E3]; in the low-R regime trained ASISR networks exceed the linear oracle in every cell while differing among themselves by at most 0.03 in 10 of 12 cells [src: E3], which describes similarity in these settings; and the benefit of the cell/footprint readout is negligible on average in the high-R band and in these evaluated seed1 cells appears where the network has partially recovered the band—consistently in LTE and SC-INR (cross-architecture Spearman 0.985, seed 1) [src: F1]; averaging the full model over training scales predicts the ranking of these losses after the fact (Spearman 0.700–0.954 over four completed seed2/3 groups; correlational) [src: L1L2].

4. **A prospective test of practical guidance** (§5.6). Independent-calibration risk selection improves a fixed GP default on a fully frozen 36-case cohort, but selects the same spectrum throughout and is less accurate and more expensive than frame-holdout CV. The test separates a useful calibrated prior from a demonstrated advantage of target-specific analytical selection [src: G2-result; G2-audit].

Scope and honesty. Props. 2–3 follow directly from classical Wiener/LMMSE theory; our claim is not new estimation theory but its use as an a-priori diagnostic and prescription for spectral adaptation in continuous representations, together with a pre-registered test across tasks. Several pre-registered criteria failed; we report them in the main text and in §6.

Completed replication and scope. Seed2/3 tests support a simple footprint predictor in four groups and L2 magnitude in three. Function similarity fails in all four, and L2 superiority over the simple predictor fails the 3/4-group criterion. Classical identities, oracle diagnosis and prospective selection are different claims. G2 supports improvement over its specified default, but not adaptive superiority or savings over CV.

2 Related Work

Spectral control in implicit representations. SIREN [Sitzmann et al., 2020] sets the frequency scale ω_0 by hand; Fourier features [Tancik et al., 2020] expose the bandwidth σ and connect it to the neural tangent kernel [Jacot et al., 2018]; FINER [Liu et al., 2024] widens the reachable frequency set. A recent group selects these parameters from the spectrum of the target signal: FreSh [Kania et al., 2024] matches the spectrum of the untrained model to the target's; WINNER [Chandravamsi et al., 2025] sets Gaussian weight-perturbation scales from the target spectral centroid; SEC [Dądela et al., 2026] retrieves grid-search-derived frequency parameters using centroid-based calibration. WIEN-INR [Ni et al., 2026] uses wavelet-band subnetworks and a finest-detail enhancement module. These studies include signal fitting, and FreSh and WINNER also evaluate inverse tasks. We focus on observation- and noise-conditioned linear references, mismatch costs, and the empirical reliability of guidance under degraded observations. Spectral bias [Rahaman et al., 2019] explains what an untuned network prefers; our budget explains what the observation permits.

Arbitrary-scale super-resolution. LIIF [Chen et al., 2021] decodes a local latent at continuous coordinates with a cell (footprint) input; LTE [Lee and Jin, 2022] adds a local dominant frequency estimator; MetaSR [Hu et al., 2019], SRNO [Wei and Zhang, 2023], CiaoSR [Cao et al., 2023] and HIIF [Jiang et al., 2025] refine the decoder; the SC-INR research implementation examined here replaces the cell input with an analytic footprint (sinc) readout and includes scale-consistency components; their benefits are evaluated separately from the footprint. Gaussian-splatting decoders [Chen et al., 2025] and heat-field decoders with analytic anti-aliasing [Becker et al., 2025] form a newer family. A controlled re-training study [Nasir et al., 2026] reports architectural saturation of the MLP/operator family. Low-pass behaviour of SR networks has been measured with a global scalar [Deng et al., 2024]; we measure recovery per band defined by the sampling theorem and set it against a covariance-conditioned linear reference. We examine these empirical similarities and their limitations.

Multi-frame and sparse reconstruction. SuperF [Jyhne et al., 2025] fits a Fourier-feature network to multiple shifted low-resolution frames and tunes σ; Deep Image Prior [Ulyanov et al., 2018] reconstructs without training data; kriging/GP regression [Rasmussen and Williams, 2006] inserts the spectrum directly as a prior. Our map conditions their possible benefit on the observation model and noise, rather than assigning a method family a fixed recoverability regime.

Classical theory and per-band diagnostics. LMMSE/Wiener estimation and aliasing analysis of sampling are textbook [Kay, 1993]. Per-frequency agreement measures such as Fourier ring correlation in microscopy [van Heel and Schatz, 2005] and scale-decomposed skill scores in forecast verification are close relatives of our per-band recoverability; to our knowledge none has been used to predict when spectral adaptation of continuous representations helps.

3 Spectral Adaptation as Choosing a Prior

Setting. A signal x lives on a continuous domain $\Omega \subset \mathbb{R}^d$. We observe $y = Ax + n$ with a known linear operator A (blur and decimation, sparse sampling, or the footprint integrals of several shifted frames) and $n \sim \mathcal{N}(0, \sigma_n^2 I)$. A continuous representation f_θ is fitted by minimising $\|Af_\theta - y\|^2$ (possibly with regularisation or early stopping) and queried anywhere.

Proposition 1 (Fourier features are a GP prior; Bochner)**.** *Let $f(u) = \sum_{j=1}^m w_j \varphi_j(u)$ with $\varphi_j(u) = \sqrt{2/m}\cos(2\pi b_j^\top u + c_j)$, $b_j \sim p(b)$, $c_j \sim \mathcal{U}[0, 2\pi)$ and $w \sim \mathcal{N}(0, s^2 I)$. As $m \to \infty$, f is a zero-mean stationary Gaussian process with covariance $k(h) = s^2\, \mathbb{E}_b[\cos(2\pi b^\top h)]$ and spectral density $S(\nu) = s^2[p(\nu) + p(-\nu)]/2$, equal to $s^2 p(\nu)$ for symmetric p.*

Hence a ridge read-out of Fourier features is the GP posterior mean under prior spectrum S; with $p = \mathcal{N}(0, \sigma^2 I)$, $S(\nu) \propto \exp(-\|\nu\|^2/2\sigma^2)$. Tuning σ (SuperF) or choosing frequency parameters from a target spectrum (FreSh, WINNER, SEC) thus picks a prior spectrum within a restricted family; a GP with a fitted or known spectrum picks it directly. For an MLP on Fourier features trained by gradient descent, the lazy-training limit gives kernel regression with a stationary NTK modulated by p [Tancik et al., 2020, Jacot et al., 2018], under additional architectural and optimisation assumptions. General trained MLPs and grid fields do not thereby have known stationary spectral priors; we compare their behaviour empirically (§5.4).

Proposition 2 (Linear upper bound)**.** *For zero-mean or correctly centred signal and noise, let x have covariance C_P, uncorrelated noise covariance $\sigma_n^2 I$, and positive-definite C_y. Every linear estimator $\hat{x} = Gy$ satisfies $\mathbb{E}\|x - \hat{x}\|^2 \geq \mathbb{E}\|x - \hat{x}_{\mathrm{LMMSE}}(P)\|^2$, with $\hat{x}_{\mathrm{LMMSE}}(P) = C_P A^\top (A C_P A^\top + \sigma_n^2 I)^{-1} y$.*

This optimum holds in expectation under the specified covariance and noise. Second moments and zero signal–noise cross-covariance suffice for linear optimality; joint Gaussianity additionally identifies the conditional mean with MMSE. A generic spectrum does not define a distribution-free ceiling on real images, and an image periodogram is an oracle reference rather than an attained optimality certificate. For any fixed linear H and evaluation map W, with $C_y = AC_PA^\top + C_n$ positive definite,

$$\begin{aligned}&\mathbb{E}\|W(x - Hy)\|^2 - \mathbb{E}\|W(x - H_P y)\|^2 \\ &= \mathrm{tr}[W(H - H_P)C_y(H - H_P)^\top W^\top] \geq 0,\end{aligned}$$

where $H_P = C_P A^\top C_y^{-1}$. The same covariance-conditioned optimum minimises total and projected linear risk.

Definition 1 (Per-band linear recoverability)**.** For a band projector Π_b,

$$R_b^{\mathrm{lin}}(A, \sigma_n, P) = 1 - \frac{\mathbb{E}\|\Pi_b(x - \hat{x}_{\mathrm{LMMSE}}(P))\|^2}{\mathbb{E}\|\Pi_b x\|^2}.$$

R_b^{lin} depends only on A, σ_n and P. With the generic natural-image prior $P \propto 1/f^2$ it requires no ground truth and no training; we compute it by sampling from the prior, simulating the observation and applying the LMMSE, with explicit reference-sampling and solve costs. Its amplitude is an additional prior assumption. When a trained or otherwise non-linear estimator is evaluated on real images, we report the same quantity measured on data, $R_b = 1 - \|\Pi_b(x - \hat{x})\|^2 / \|\Pi_b(x - \bar{x})\|^2$ using the colour aggregation specified by each protocol; for a trained network we write R_b^{net}.

Bands. Image bands are square annuli in HR pixel units, indexed by $\max(|\nu_x|, |\nu_y|)$, $b_1 = [0, \frac{1}{8})$, $b_2 = [\frac{1}{8}, \frac{1}{4})$, $b_3 = [\frac{1}{4}, \frac{1}{2}]$ cycles/pixel. At $\times 4$, b_1 lies below the LR Nyquist limit and b_2, b_3 lie one and two octaves above it.

Closed form for decimation. For ideal periodic $A =$ blur K followed by decimation by s, each LR frequency ω collects s^d HR frequencies $\nu_m = \omega + m/s$, and

$$R(\nu_0) = \frac{|\hat{K}(\nu_0)|^2 P(\nu_0)}{\sum_m |\hat{K}(\nu_m)|^2 P(\nu_m) + N_\omega}. \tag{1}$$

The sum in the denominator is aliasing. In single-image and sparse-pixel settings it dominates the sensor noise, which is why R_b^{lin} there is nearly insensitive to σ_n (Table 1).

Here N_ω is noise power under a compatible Fourier normalisation. Changing other powers within the alias set changes the effective noise; the scalar mismatch identity assumes that effective noise is fixed. Finite crops, boundary interpolation, rotations and sparse sampling normally require the matrix risk expression. F1/G1 use HR256 and 73 images; L1/L2 use HR240 and 74, with their own window and pooling conventions. Empirical recovery can be negative and need not equal reference recovery.

Zero-noise singular cases require a pseudoinverse or explicitly stated regularisation. A numerical noise floor changes the estimator; its value must not replace the physical observation-noise covariance in risk evaluation.

4 Mismatch Cost and Three Regimes

Consider a decoupled scalar channel with zero signal–noise covariance, correctly specified observation gain and unchanged effective noise. Let signal variance p and effective noise q—sensor noise plus aliased energy—so

that $R = p/(p+q)$. An estimator that assumes variance κp instead of p applies the gain $g = \kappa R/(\kappa R + 1 - R)$.

Proposition 3 (Mismatch cost)**.** *The excess mean-squared error, normalised by the scalar signal energy p, is*

$$\Delta(R, \kappa) = \frac{R(1-R)^2(\kappa-1)^2}{(1-R+\kappa R)^2}. \tag{2}$$

A proof is in Appendix A. Two limits carry the prescription:

- under-estimation, $\kappa \to 0$: $\Delta \to R$; harmless when R is low, costly when R is high;
- over-estimation, $\kappa \to \infty$: $\Delta \to (1-R)^2/R$; harmless when R is high, and larger than the band energy itself when $R < 0.38$, This compares excess risk to signal energy; total risk exceeds zero reconstruction when $\Delta > R$.

In absolute units, with $Q = \kappa P$ the assumed and P the true energy and N the effective noise at one frequency,

$$P\,\Delta = \frac{N^2(Q-P)^2}{(P+N)(Q+N)^2}. \tag{3}$$

In the well-separated over-estimation limit $P \ll N \ll Q$ the excess error is $\approx N$, independent of the signal energy; in the well-separated under-estimation limit $Q \ll N \ll P$ it is $\approx P$. Over-estimation is paid in units of noise, under-estimation in units of signal [src: DER§1.2].

Corollary 1 (Where the asymmetry flips)**.** *In the per-frequency scalar model, for every $\kappa > 1$, $\Delta(R, \kappa) > \Delta(R, 1/\kappa)$ if and only if $R < 1/2$.*

The proof follows from the sign of $(\kappa - 1)(1 - 2R)$ (Appendix A). The statement is exact under two conditions: it is evaluated per frequency, and the mismatch is a symmetric multiplicative factor (κ vs. $1/\kappa$). Averaging R and κ before applying the nonlinear formula can change the result in either direction; no universal Jensen upper or lower bound applies. Band-level crossover statements are approximate (§5.1). The effective noise is held fixed in the scalar model. Random sparse sampling couples modes, and changing several assumed powers within an alias set does not preserve a one-channel approximation. This differs from claiming that physical noise depends on the assumed prior. The sparse 10% b_1 counterexample belongs to this coupled setting.

Three regimes. We fix two thresholds before the experiments, 0.8 and 0.1 (sensitivity in §6):

- **Regime A** ($R_b^{\text{lin}} \geq 0.8$): the information is in the observation. In the limit $R \to 1$ over-estimating the spectrum costs little and under-estimating it costs much, which favours wide, low-bias representations; the bottleneck is then capacity and optimisation. This is a per-band tendency, not a rule for choosing a representation: at $R = 0.8$ a 16× over-estimate is not free, and empirical ranking uses the approximate energy-weighted cost summed over *all* bands, each with its own κ_b (Eq. (4)).
- **Regime B** ($0.1 \leq R_b^{\text{lin}} < 0.8$): both directions cost; the gain of spectral adaptation is bounded by $\text{MSE}_{\text{mismatched}} - \text{MSE}_{\text{matched}} \geq 0$, under that same true-covariance reference; matching can matter, but closeness to an oracle does not establish practical selection.
- **Regime C** ($R_b^{\text{lin}} < 0.1$): the chosen stationary reference has low expected recovery. Strong under-estimation incurs at most R excess; arbitrary over-estimation need not be safe. Alternative structured priors, including nonstationary linear models or learned nonlinear models, may recover more on real signals. This reference regime is not an impossibility theorem.

Zone labels do not pick winners. To compare representations with effective priors Q we use the band-summed, absolute cost

$$C(Q) = \sum_b E_b\,\Delta\big(R_b, \kappa_b(Q)\big), \tag{4}$$

with band energy E_b, matched-prior recoverability R_b and $\kappa_b(Q)$ the ratio of candidate to reference prior-spectrum band integrals; lower C is predicted better. In D11, the spectrum ratio is on the 512 grid, while E_b is truth energy after HR240 cropping and a Hann window. Their numerical definitions differ; the substitution is a diagnostic approximation. For linear arms the exact trace formula $\mathbb{E}\|x - \hat{x}\|^2 = \text{tr}\big((I - H_Q A) C_P (I - H_Q A)^\top\big) + \sigma_n^2 \|H_Q\|_F^2$ needs no approximation. Eq. (4) ranks representations (E2: 0.90; D11: 0.856) but does not reliably select the best among close candidates (D11: 1/3), does not cover nonlinear iterative fits, and its a-priori version (generic spectrum for E_b, κ_b) has not been tested (§5.4) [src: DER§2.3]. The per-frequency scalar form Eq. (3) illustrates why over-estimation is paid in units of noise but is not used as evidence: for non-diagonal operators (random shifts, sparse sampling) its per-frequency predictions are inaccurate.

Qualitative interpretations to test. (i) Grid or hash fields without spectral decay can express high frequencies with weak regularisation; they can deteriorate under noise, but no equivalent fixed prior Q has been identified here; (ii) Fourier features with large σ in SuperF-style fitting can show similar empirical deterioration; (iii) a regulariser of fixed strength corresponds to a fixed κ relative to a noise-dependent p/q, so its effective mismatch drifts with noise. §5.4 tests these on multi-frame SR.

Table 1: A-priori three-regime map: R_b^{lin} per band under the generic $1/f^2$ prior, no data, no training [src: OUT3/d8]. Shading: A ≥ 0.8, B, C < 0.1.

Task	σ_n	b_1	b_2	b_3
Single image ×2	.002 / .01	1.00 / 1.00	.92 / .90	.09 / .08
Single image ×4	.002 / .01	.98 / .98	.09 / .08	.01 / .01
Single image ×8	.002 / .01	.79 / .77	.01 / .01	.01 / .01
Sparse 5%	.002	.70	.09	.03
Sparse 10%	.002	.81	.20	.07
Sparse 25%	.002	.92	.47	.20
Sparse 50%	.002	.98	.76	.46
Multi-frame ×4, T=4	.002→.02	≈1.0	.73→.51	.12→.03
Multi-frame ×4, T=8	.002→.02	≈1.0	.83→.62	.27→.08
Multi-frame ×4, T=16	.002→.02	≈1.0	.90→.73	.41→.15

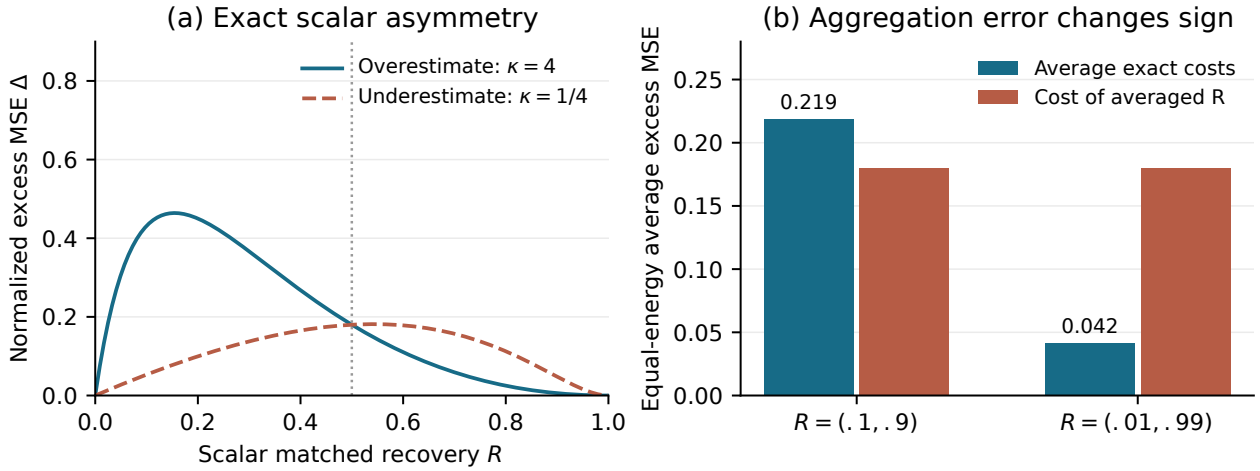


Figure 1: Exact scalar mismatch and limits of band averaging. (a) Reciprocal mismatches cross at $R = .5$ under fixed effective noise. (b) Equal-energy modes with the same average $R = .5$ and $\kappa = 4$ produce aggregation errors of opposite sign. This analytic figure does not imply scalar decoupling for the experimental observation operators.

Two ways to assign a regime. R_b^{lin} describes what the observation permits a linear estimator; it is the right axis for linear, multi-frame, sparse and 1D tasks, and it is computable in advance. A trained ASISR network, however, lifts regime-C bands to intermediate recovery through its learned prior (§5.2). To describe what components *inside* such a network do, the regime has to be assigned by the network's actual recovery R_b^{net}, which requires ground truth and is therefore a post-hoc description, not a prediction (§5.5). At this network layer Prop. 3 is not a theorem: a network is not a fixed linear Wiener estimator with a known κ. Nor do $R_b^{\text{net}} \approx 0$ or 1 force small ablation loss: removing a component can worsen a perfectly recovered band, or improve a poorly recovered one. The measured concentration of losses is therefore empirical, not a necessary boundary condition. No scalar asymmetry or causal mechanism is inferred from these network measurements.

5 Experiments

Protocol. Confirmatory thresholds follow the recovered pre-registrations. Freezing is audited experiment by experiment: E2 windowing changed after a one-image pilot, G1 predictors were proposed after seed1, and exploratory analyses are not promoted to confirmation. Images are from DIV2K validation (0801–0812) unless stated; R_b uses each recovered protocol: F1/G1/E3 average RGB to grayscale, L2 pools RGB channel energies, and E1/D11 retain RGB metrics. Exploratory analyses are labelled as such.

5.1 The mismatch cost is predictable (E2)

Setup. 12 DIV2K images, five tasks (single image ×2, ×4; sparse pixels 10%, 25%, 50%). In each of the three bands we multiply the true spectrum by $\kappa \in \{1/16, 1/4, 4, 16\}$, reconstruct with a GP (linear), and measure the lost recoverability in that band, giving

Table 2: Original E2 reported criteria (60 cells, 12-image means) [src: E2§2; E4-T1].

Criterion	Threshold	Result
a. Spearman(band-avg. prediction, measured)	≥ 0.8	**0.90**
b. $R < 0.3$: over-est. > under-est.	$\geq 80\%$	4/4
b. $R > 0.7$: under-est. > over-est.	$\geq 80\%$	6/7
c. median |pred.−meas.|	≤ 0.05	0.038
d. Spearman(a-priori prediction, measured)	≥ 0.7	0.79

Table 3: Original E2 measured / band-averaged predicted cost Δ (fraction of band energy) [src: E2§3].

Task, band	R	κ=$\frac{1}{16}$	$\frac{1}{4}$	4	16
SI×2 b_2	.80	.230/.402	.047/.101	.067/.031	.165/.056
SI×2 b_3	.11	.084/.097	.044/.058	.194/.380	**1.370**/2.648
SI×4 b_2	.085	.060/.074	.029/.045	.150/.336	**0.933**/2.590
Sparse 10% b_2	.37	.197/.291	.072/.144	.110/.302	.586/.976
Sparse 25% b_2	.70	.226/.430	.068/.139	.094/.074	.473/.149
Sparse 50% b_2	.90	.148/.264	.036/.046	.024/.009	.064/.015

60 cells [src: E2§2]. Band energies are computed after a Hann window on both signal and error; the window was not specified in the pre-registration and was added after a one-image run showed boundary leakage (single-image ×4 b_3 measured $R = 0.56$ at 0.004% transmittance). After the fix, single-image ×4 b_2 measures $R = 0.085$ against an a-priori 0.08 [src: E2§1]. Only the metric changed, not hypotheses or criteria.

Independent reimplementation (E2r). A separate implementation reproduces all 60 aggregated cells from 12 images and five tasks. Its post-hoc cost ranking is .902528, reference-prior ranking .797188 and median absolute error .035099 (maximum 1.8895). Asymmetry counts are 4/4 and 6/7; all four frozen thresholds pass. These are distinct rerun numbers, not replacements for individual original cells in Tables 2 and 3. The high maximum error limits quantitative interpretation despite the passed median threshold [src: E2r].

Results. All four pre-registered criteria hold (Table 2). In low-R bands under-estimation is nearly free (≤ 0.06) while a 16× over-estimate costs 0.9–1.4 times the band energy, i.e. worse than not reconstructing the band (Table 3). In high-R bands the direction reverses. When $R \approx 0$ (single image ×4 b_3) neither direction costs anything, because the observation carries nothing about the band [src: E2§3]. The band-averaged prediction over-estimates large-κ costs in the reported examples; this direction is not general. For decimation, evaluating Eq. (2) per frequency with Eq. (1) and energy-weighting reduces approximation error in the reported exploratory calculation (exploratory: Spearman 0.99, median error 0.004; e.g. SI×4 b_2, $\kappa = 4$: predicted 0.151, measured 0.150) [src: E2§4]. The single counter-example to the asymmetry is sparse 10% b_1 ($R = 0.92$; under 0.072 vs. over 0.110): random sampling spreads aliasing across bands, outside the scalar model [src: E2§2].

Table 4: Single image ×4, R_b measured on data [src: OUT7.1/h6c]. Regime by R_b^{lin}: b_2, b_3 are C. gen./orac.: LMMSE with generic/oracle spectrum; SC-eq/SC-ns: SC-INR-eq / SC-INR-nosinc.

Set	band	gen.	orac.	LIIF	LTE	SC-eq	SC-ns
Set14	b_2	.16	.25	.56	.57	.57	.56
	b_3	.09	.09	.20	.20	.20	.18
BSD100	b_2	.13	.22	.38	.38	.38	.38
	b_3	.06	.06	.13	.13	.13	.11
Urban100	b_2	.11	.27	.50	.50	.51	.51
	b_3	.02	.03	.14	.14	.15	.13

5.2 Low-reference-recovery bands: similarities among evaluated ASISR decoders (M2, E3)

Setup. We compare four trained decoders that share the EDSR-baseline encoder and the LIIF training protocol (DIV2K, 1000 epochs, scale range ×1–×4, seed 1): LIIF, LTE, SC-INR-eq and SC-INR without the sinc footprint readout. This early comparison is not a matched SC-INR/NoSinc ablation; the latter is evaluated separately in F1/G1 against two linear references, LMMSE with the generic spectrum and LMMSE with the image's own (oracle) spectrum, on 256^2 centre crops of the first 30 images of Set14, BSD100 and Urban100 [src: OUT7.1/h6c].

×4. In both regime-C bands the trained networks recover more than the evaluated oracle-periodogram linear reference: 0.38–0.57 vs. 0.22–0.27 in b_2 (1.7–2.3×) and 0.13–0.20 vs. 0.03–0.09 in b_3 (2–6×) (Table 4). This comparison is consistent with benefits from a learned prior, but does not prove that all linear models or all alternative priors fail on these images. The four decoders differ by at most 0.02 in either band.

Across scale factors (E3). We repeat the measurement at ×2, ×3, ×8 (27 cells: 3 scales × 3 bands × 3 datasets) with the same networks; ×8 is out of the training range [src: E3]. Pre-registered results (Table 5): the a-priori R_b^{lin} ranks the measured generic-LMMSE recovery with Spearman 0.952; in 10/12 regime-C cells all four networks exceed the oracle LMMSE and differ by at most 0.03; in 9/9 regime-A cells the best network is within 0.05 of the oracle LMMSE (largest gap +0.016); generic-LMMSE recovery decreases monotonically with scale in 9/9 band–dataset pairs. The zones move as predicted: b_2 is A at ×2 ($R_b^{\text{lin}} = 0.92$), B at ×3 (0.42) and C at ×8 (0.011); b_1 enters B at ×8 (0.79) [src: E3-key]. In all 12 C cells every network exceeds the oracle (by +0.002 to +0.244) [src: E4-T2].

Table 5: E3 pre-registered criteria [src: E3; E4-T1].

Criterion	Threshold	Result
a. Spearman(R_b^{lin}, generic LMMSE)	≥ 0.8	0.952
b. C: nets > oracle, range ≤ 0.03	$\geq 80\%$ cells	10/12
c. A: best net − oracle ≤ 0.05	$\geq 80\%$ cells	9/9
d. generic LMMSE monotone in scale	9/9	9/9
e. (expl.) oracle−generic: B>A, B>C	–	.078 vs .011/.040

Scale collapse (L1, pre-registered). If regime C is set by the observation rather than by the decoder, recovery should depend on frequency relative to the LR Nyquist limit, $u = 2s\nu$, not on absolute frequency. On 240^2 crops of the first 30 images per dataset, we measure $R(u)$ in seven u bins at $\times 2/\times 3/\times 4$ for LIIF, LTE and SC-INR (seed 1) and take the median range across scales, D_{in} [src: L1L2]. The linear control collapses (D_{in} 0.019 / 0.006 / 0.011 on Set14 / BSD100 / Urban100). Binning by u instead of ν shrinks the range by an order of magnitude in 9/9 model–dataset groups (0.013–0.061 vs. 0.31–0.46). The main criterion, $D_{\text{in}} \leq 0.05$ in $\geq 8/9$ groups, fails: 6/9, with all three models on Set14 at 0.055–0.061 (BSD100 and Urban100: 0.013–0.019). This exceeds both the seed noise floor (median range across seeds 0.003–0.004) and the linear control; the cause is not diagnosed. We found no consistent deviation outside the training range ($\times 6/8/12$; $D_{\text{out}} > D_{\text{in}}$ in only 6/9), and the public models generalise only partly (GSASR and Thera pass, ContinuousSR fails; 6/9) [src: L1L2]. Above the LR Nyquist limit, our three decoders exceed the generic linear estimate by +0.18 to +0.40 and show small differences in these evaluated cells [src: L1L2].

Reading. These evaluated networks have similar band recovery in 10/12 low-reference-recovery cells at a .03 range threshold, but only 8/12 at .02. This does not establish a universal limit on decoder design, and changing the learned prior or training recipe remains outside this comparison. The oracle-periodogram estimate is a reference rather than a certificate of optimal realised linear recovery.

5.3 Spectral-prior effects under sparse observations (M3, M4)

Sparse pixels. Sampling rates 5–50%, DIV2K 0801–0812, luminance; we compare linear estimation with the generic spectrum, with the image's own spectrum, and DIP [Ulyanov et al., 2018] [src: OUT7.2/sp_scan]. The true-spectrum linear estimator beats DIP at every rate (26.39 vs. 25.19, 28.45 vs. 27.02, 32.66 vs. 30.84, 37.89 vs. 34.79 dB at 5/10/25/50%), and the generic spectrum trails it by 1.8–4.5 dB (Table 6). The space between the two linear estimators is the observed gap between these two evaluated linear references.

Table 6: Sparse-pixel reconstruction: R_{b_2}, R_{b_3} and PSNR (dB) for generic / true-spectrum linear / DIP [src: OUT7.2].

Rate	b_2	b_3	PSNR
5%	.17/.39/.16	.10/.20/.05	24.59/26.39/25.19
10%	.29/.56/.34	.14/.31/.12	26.13/28.45/27.02
25%	.59/.81/.69	.29/.57/.38	29.27/32.66/30.84
50%	.85/.95/.89	.57/.82/.65	33.37/37.89/34.79

Table 7: E1: 12-image mean PSNR (dB), $T = 16$ [src: E1§2, e1_result_v2.json].

Arm	σ_n=0	.005	.01	.02
mrgrid	31.39	28.36	22.57	14.36
ls_tv1e-4	31.16	27.37	21.24	13.78
ls_tv1e-3	29.98	29.61	28.64	24.84
ff_3	27.94	27.92	27.88	27.74
ff_10	28.86	28.79	28.61	28.01
gp_gauss_30	**31.62**	27.73	26.65	25.56
gp_power_2.0	31.19	29.37	27.37	26.10
gp_power_2.5	29.62	29.36	28.65	27.45
gp_power_3.0	28.27	28.21	28.04	27.50
gp_oracle	29.91	**29.70**	**29.25**	**28.35**
bilinear	24.65	24.64	24.59	24.41

1D EEG. On 200 calibration segments of EEGMMIDB [Schalk, 2009, Schalk et al., 2004] with 10/25/50% of samples kept, the spectral prior helps most in θ/α at 10–25% and almost nowhere at 50%, where most bands are already A; e.g. α at 10%: linear interpolation .22, population-average spectrum .30, true spectrum .47 [src: OUT7.3]. These are calibration-only numbers from the original report. The 760-segment test result has not been recovered and is not counted as confirmatory evidence here.

5.4 Multi-frame SR across noise levels (E1)

Setup. DIV2K 0801–0812, $\times 4$, $T = 16$ shifted frames synthesised neutrally (footprints integrated on the 512^2 original, ground truth by 2×2 averaging), known registration, RGB, PSNR with an 8-pixel border crop; $\sigma_n \in \{0, 0.005, 0.01, 0.02\}$ [src: E1]. Arms: a multi-resolution grid field (mrgrid); least squares with TV (ls_tv); a SuperF-style Fourier-feature MLP (ff_σ); GPs with Gaussian spectra (gp_gauss_σ), power-law spectra (gp_power_α) and the image's own spectrum (gp_oracle); bilinear. Paired bootstrap 95% CIs over images.

Results. The direction predicted by Prop. 3 holds: representations without spectral decay (mrgrid, weak TV) collapse under noise, losing 17 dB between $\sigma_n = 0$ and 0.02 (Table 7), the over-estimation failure; the value of the true spectrum over the best power-law GP grows

with noise, D from 0.33 to 0.85 dB, difference +0.51 [0.13, 0.88] (P3 holds); and the winner moves from a low-bias arm (gp_gauss_30) to gp_oracle (P4 holds) [src: E1§1]. At $\sigma_n = 0$ a low-bias GP beats the best Fourier-feature MLP by +2.76 [1.99, 3.56] dB.

Two pre-registered predictions fail [src: E1§1]. (P1) At $\sigma_n = 0.005$ low-bias representations no longer win: mrgrid − ff_10 = −0.43 [−1.01, 0.14] dB; the switch happens earlier than predicted. (P2) At $\sigma_n = 0.02$ the generic power-law GP does not come within 0.6 dB of the oracle (−0.85 [−1.10, −0.61]). The winner margins at $\sigma_n = 0$ and 0.005 are not significant (+0.19 and +0.10 dB, CIs include 0).

Why the switch came early. The pre-registered prediction used one zone label: at $T = 16$, $R^{\text{lin}}_{b_2}$ falls from 0.90 to 0.73 over the noise range (Table 1), so b_2 is regime A at small noise and low-bias arms were expected to win. It ignored each arm's mismatch κ_b and ignored b_3. The D11 diagnostic below shows that both matter. Two further notes. P4's switch at 0.005 partly reflects gp_oracle's numerical noise floor at $\sigma_n = 0$ (it is an oracle reference; covariance-conditioned optimality does not certify its realised per-image performance), so future switch predictions should compare practical arms only [src: DER§2.2]. For the non-linear arms, mrgrid and ls_tv1e-4 already lose more at $\sigma_n = 0.005$ than their noiseless lead over ff_10, which is the direct numerical reason P1 fails; their mechanism was not tested [src: DER§2.2].

Diagnostic on E1 (D11, pre-registered). We stored per-frequency error spectra of the GP arms on the E1 images and settings (CPU; the non-linear arms mrgrid, ls_tv and ff were not measured) [src: D11]. At $\sigma_n = 0.005$ the excess error of gp_gauss_30 over gp_oracle splits 0.5% / 44% / 56% over $b_1/b_2/b_3$. In b_3 the measured oracle recovery is $R = 0.22$ (regime B) with over-estimation $\kappa = 4.5$; in b_2, $R = 0.80$ but $\kappa = 16.4$ (Table 8). The pre-registered rule looked only at b_2; b_3 was already in regime B, and in b_2 a large κ is not free even at $R = 0.8$, since the "harmless over-estimation" of regime A is the $R \to 1$ limit. Pre-registered results: (a) the b_3 share of gp_gauss_30's excess is 0.556 [0.464, 0.618], above 0.5 at the point estimate but with a CI that includes it; (b) the band-wise predicted excess $\sum_b E_b\, \Delta(R_b, \kappa_b)$ ranks the measured excess over 24 arm–noise points with Spearman 0.856 [0.741, 0.918] (point threshold 0.8 holds, but the CI includes lower values); (c) it does *not* pick the best non-oracle GP arm (1/3 noise levels, threshold 2/3): the top arms are close and the band-averaged form carries a band-aggregation error (e.g. gp_power_2.0 at $\sigma_n = 0.005$: predicted 1.40×10^6, measured 3.9×10^4, unnormalised) [src: D11]. The concentration of excess error on $R(\nu) < 1/2$ frequencies (0.82 / 0.71 for the two wide arms vs. 0.18 for gp_power_3.0) has no discriminative value, because 78% [74%, 81%] of all frequencies satisfy $R(\nu) < 1/2$ at this noise level; we do not use it as evidence. So the band-wise cost $C(Q)$ (Eq. (4)) ranks the GP arms but does not select the winner; whether its a-priori version ranks equally well has not been tested [src: DER§2.3].

Table 8: D11: gp_gauss_30 vs. gp_oracle, E1 multi-frame, $T = 16$, $\sigma_n = 0.005$, 12-image mean [src: D11].

Band	oracle R_b	κ_b	share of excess
b_1	1.00	1.02	0.5%
b_2	0.80	16.4	44%
b_3	0.22	4.5	56%

Bootstrap and diagnostic boundary. The saved D11 script used 2,000 source-image resamples. We reproduced those draws and extended the same paired procedure to the protocol's 10,000. The updated ranking CI is [.741,.918]; an independent 10,000-draw image-cluster calculation gives b_3 share [.464,.618]. Both condition on 12 images and fixed noise draws. D11 uses truth-derived energy, oracle recovery and spectrum ratios; its ranking is post-hoc and cannot establish a target-observation-only selector. The cropped/windowed empirical R_b and band ratios also do not satisfy the exact scalar model by construction [src: CPU-10000].
Non-linear arms (mrgrid, ls_tv, ff) are covered only qualitatively; $T = 4/8$ (4 images) is descriptive only.

5.5 Inside the network: SC-INR's footprint readout (F1)

SC-INR's only component with a reliable benefit in our earlier ablations is its footprint readout (analytic sinc integration over the target pixel); in LTE the analogous input is the cell. F1 asks where in the band map this component matters. We ablate it in two architecture pairs (LTE vs. LTE-NoCell; SC-INR vs. SC-INR-NoSinc) and measure the per-band loss $L = R_{\text{full}} - R_{\text{ablate}}$ on 27 confirmatory cells ($\times 2/\times 3/\times 4 \times 3$ bands $\times$ 3 datasets), seed 1 [src: F1]. The pre-registered predictor is $g(R) = R(1-R)^2$, the small-mismatch limit of Eq. (2).

Results (LTE pair). L is zero where $R \approx 1$: mean L over regime-A cells is 0.0000, versus 0.0050 at $\times 3$ b_2 (regime B) [src: F1; E4-T1]. The loss concentrates in b_3 at $\times 2$–$\times 4$ and b_2 at $\times 4$, where R^{lin}_b is only 0.01–0.10 (regime C) but the network recovers 0.13–0.56. The a-priori predictor fails: Spearman$(g(R^{\text{lin}}_b), L) = 0.411 < 0.5$ (standard Spearman with average ranks for ties). Using the network's own recovery raises it to 0.490, still below the threshold [src: E4-T1].

Results (SC-INR pair). With the matched pair (SC-INR vs. SC-INR-NoSinc, same encoder), the picture is

Table 9: Completed G1 confirmation; 27 aggregated cells per group.

Group	$\rho(\epsilon_b^2, L)$	$\rho(\epsilon_b^2 h(R), L)$	$L_3 > L_2$
LTE seed2	.577	.532	8/9
SC seed2	.943	.703	9/9
LTE seed3	.676	.643	8/9
SC seed3	.885	.804	9/9

Table 10: L2 replication. Ratio is median L_{pred}/L for $L > .005$; similarity is the fraction closer to marginal than full output.

Group	ρ	ratio	similarity	$\rho - \rho_\epsilon$
LTE seed2	.712	.823	.770	+.130
SC seed2	.755	.581	.689	−.142
LTE seed3	.954	.833	.748	+.331
SC seed3	.700	.337	.653	−.173

the same: the A-regime criterion holds, the a-priori predictor fails (Spearman$(g(R_b^{\text{lin}}), L) = 0.400$), the network-recovery predictor also stays below threshold (0.492), and the per-cell losses of the two architectures agree closely (Spearman 0.985) [src: F1-corr2]. Mean L in the A cells is -0.000609, versus 0.001034 at $\times 3$ b_2, from the saved corrected table. All F1 numbers are seed 1 only; seeds 2/3 are evaluated under the G1 pre-registration below.

A closed-form band-order predictor (G1, completed). The box-footprint versus point-sampling quantity $\epsilon(\nu) = 1 - \text{sinc}(\nu_x)\,\text{sinc}(\nu_y)$ provides three fixed band scores (.00006, .00552, .07235). It was chosen after seed1; only seed2/3 count toward confirmation. The original protocol was recovered byte-for-byte and matched its frozen SHA256 (`548b26fe` prefix). Correctly paired HR256 evaluations contain 13 Set14, 30 BSD100 and 30 Urban100 images; $\times 3$ uses LR85. H-G1 passes in 4/4 groups (at least 3 require $\rho \geq .5$); H-G2 passes in 34/36 settings (threshold 26); H-G3 passes in 4/4: $h(R) = R(1-2R)^2$ adds no improvement within the .05 allowance. This supports coarse frequency-band ordering, not a quantitative gain mechanism. With only three distinct predictor values, tied average ranks are essential. The linearised gain account was separately rejected by L2b. Newly computed predictions were not used to tune the thresholds [src: G1-20261008].

Predicting the loss from the full model (L2, pre-registered). One account is that the cell or footprint input tells the decoder the scale factor, so a model without it behaves like the full model averaged over the training scales. We test this without training: average the full model's output over cells corresponding to $s' \in \{1.0, 1.1, \ldots, 4.0\}$ and predict the ablation loss as $L_{\text{pred}} = R_{\text{full}} - R_{\text{marg}}$ on the 27 F1 cells, seed 1 [src: L1L2]. The ranking matches: Spearman$(L_{\text{pred}}, L) = 0.822$ (LTE) and 0.824 (SC-INR), against 0.40–0.49 for the zone predictors of F1 and $+0.147/+0.149$ above the footprint gain ϵ_b^2 (0.676 on this evaluation) [src: L1L2]; with log-uniform s' it is 0.805/0.803. Two pre-registered criteria fail. The magnitude matches only at $\times 4$ and above: the median L_{pred}/L is 0.59 (LTE) and 0.36 (SC-INR), outside $[0.5, 2]$ for SC-INR, because at $\times 2/\times 3$ the ablation loses more than marginalisation predicts (e.g. BSD100 $\times 3$ b_3: 0.038 vs. 0.008). The ablated output is closer to the marginalised than to the full output in 73.9% / 69.4% of image–scale pairs (threshold 80%), rising with scale (LTE: 58% at $\times 2$, 99% at $\times 8$) [src: L1L2]. So the ablation loss can be predicted after the fact from the full model's cell average, consistent with the readout supplying scale information at large scale factors; at small scale factors the ablation loses more than that, and no per-frequency linear account reproduces it [src: DER§3.4]. This is correlational evidence; a causal test needs a pure scalar-s embedding arm, which requires training and has not been run.

Completed L2 seed2/3 replication. All four frozen groups are now evaluated at HR240 (14/30/30 images), with 31 linear and 31 log-spaced cell queries. The primary 27 cells exclude $\times 8$. Independent recalculation from per-image signal and error energies reproduces all pooled values. Under the original at-least-3/4 rule, H2a ordering passes 4/4 and H2b magnitude passes 3/4. H2c function similarity passes 0/4 (threshold .8), and H2d superiority over ϵ_b^2 passes 2/4 (required improvement .05); both fail the at-least-3/4 hypothesis criterion. The advantage occurs only in LTE. H2z's cell-invariant ablation control passes 4/4 with maximum difference zero. G1 and L2 use different crops and pooling; their rows cannot be interchanged. These findings do not establish that removing the readout produces the scale-marginalised function, or that scale marginalisation is the best cross-architecture mechanistic account [src: L2-20261008].

5.6 Prospective guidance: independent calibration (G2)

Frozen setup. We test eight fixed GP spectra: Gaussian widths $3, 10, 30, 60, 128$ and power-law exponents $2, 2.5, 3$. The default is $1/f^2$. DIV2K training images 0001–0032 calibrate a unit stationary covariance; 0801–0812 are development images. The complete target cohort is 0813–0824 at $\sigma_n = .005, .01, .02$ (36 cases). These targets were not used to develop this selector; they are not claimed to be globally unseen in all earlier research. Matched $y = Ax + \epsilon$ uses HR256, $\times 4$, 16 registered frames, eight footprint subsamples per axis and a 512 field. RGB PSNR uses an eight-pixel crop and unclipped predictions. This controlled geometry differs from E1's native forward

model [src: G2-protocol].

Observation-only rules. All candidates share a contrast estimate from noise-corrected observation energy and independent calibration probes. Analysis selects the minimum expected cropped MSE under the calibrated covariance, estimated by four common Gaussian probes with physical noise and the observed-mean intercept. This is a plug-in reference risk, not an unbiased target-risk estimate. Four-fold CV holds out frames by index modulo four and estimates contrast from training frames only. Each method refits its chosen prior on all frames. Each selector has 32 fits, but analysis uses full frames and CV uses 12: equal fit counts do not imply equal cost. All 36 choices and chosen outputs were SHA-frozen before scoring opened target truth. Double-precision PCG permits 4000 iterations and requires a freshly evaluated relative residual below 10^{-5}. Earlier development failures are retained; no target cases were removed. All 36 completed with no failures; 2592 saved selection/evaluation solver certificates pass, with maximum 3134 iterations [src: G2-audit].

Results and counterexamples. Analysis improves the default by 1.105 dB [.726, 1.504], with no $>$ 1 dB degradations, passing the predeclared default-improvement gate (positive lower interval bound and at most 5% severe degradations). Its gains at the three noises are $1.247, 1.124, .945$ dB. CV improves the default by 1.217 dB. The paired analysis-minus-CV gain is $-.112$ dB $[-.313, -.002]$, and analysis costs 1.76 times CV. The predeclared superiority-at-no-greater-cost gate fails on both gain and cost. The frozen summarizer additionally requires zero selection failures for that gate; there are none, so this extra restriction does not change the conclusion [src: G2-result; G2-audit].

Analysis chooses $1/f^{2.5}$ in 36/36 cases, whereas CV chooses exponents 2/2.5/3 in 3/27/6 cases. They hit the eight-candidate oracle in 25/36 and 34/36 cases respectively. Analysis loses to the default on image 0814 at all three noises, worst $-.152$ dB; its largest oracle regret is 1.248 dB on 0815 at .01 noise. A retrospective fixed-$1/f^{2.5}$ replay has identical outputs to analysis, with common preparation and final fits costing 194.6 s. This algebraic/cache identity is an interpretation of the completed results, not an added confirmatory baseline or a new default chosen before this test. The experiment does not isolate a benefit of target-specific analytical selection [src: G2-audit].

Full cost and practical reading. Analysis and CV require 5.065×10^{12} and 3.222×10^{12} interpolated points. Peak recorded allocated memory is .488 GB. Calibration takes 3.0 s separately; the actual resource ledger records 14,346.6 s of GPU-phase invocation wall time, including the experiment's selection and oracle scoring, within the 24-hour budget. These timings depend on the shared-server load and are not a benchmark of exclusive-device speed. Independent CPU audits reproduce statistics and costs; all 78 cached chosen predictions were re-scored from RGB pixels. Uncached candidate predictions were not retained, so their oracle scores were audited from saved MSE and convergence records, without an independent GP rerun. In this task, calibrated analysis finds a better prior than the specified default, but frame-holdout CV adapts more effectively and costs less. We therefore retain the analysis as a conditional diagnostic, not a demonstrated cheap replacement for CV or a neural-architecture selector [src: G2-audit].

6 Limitations

- **Conditional reference.** Classical LMMSE optimality assumes specified covariance, noise and operator. A generic spectrum is not a universal bound; individual-image periodograms are oracle references.
- **Band approximation.** Scalar decoupling need not hold for aliased or rotated observations. Band averaging has no uniform error direction. Windows, cropping, clipping and solver floors create additional departures from exact theory.
- **Network mechanism.** G1 and L2 establish empirical ordering, not a gain-mismatch mechanism or function equivalence. L2 function similarity passes in 0/4 groups; superiority over the simple score passes only 2/4. No scalar-scale-only retraining arm has been evaluated. Exploratory same-observation controls in Appendix D further retain opposing kernel/FF orderings; neither finite Fourier ridge nor a model's own posterior variance establishes the trained network mechanism.
- **Failed tests and sensitivity.** F1, E1 switching, D11 winner selection and L1 strict collapse retain their failures. E3 similarity passes 10/12 at .03 and 8/12 at .02; its .1/.8 reference-regime thresholds remain descriptive pending the sensitivity audit.
- **Replication scope.** G1/L2 cover two additional fixed seeds and two pairs. G1 uses HR256 and 73 images ($\times 3$ means 256/85), while L2 uses HR240 and 74. Their pooled rows cannot be interchanged. G1 saves per-image recovery but not signal/error energies, limiting independent reconstruction of its pooled statistics.
- **Protocol and numerical differences.** E2 windowing changed after a one-image pilot. E3 has no injected observation noise although its reference uses $\sigma = .002$ and its GP solver $\nu = 10^{-5}$. E1's native forward geometry differs from its inverse discretisation, and clipped outputs are not fixed linear estimators.

Table 11: Complete G2 cohort. Gain is relative to fixed $1/f^2$; 95% intervals use 10,000 bootstrap draws of 12 image clusters, averaging three noises within each image. Regret is against the best of the eight fixed candidates scored after freezing. Costs include common contrast preparation, all selector fits and the chosen final fit, once per deployed policy; shared fits are not added twice to actual experimental spending. Calibration overhead is reported separately.

Rule	Mean gain [95% CI], dB	Regret mean/median/p90, dB	Worst gain, dB	>1 dB loss	Cost, s
Default	0	1.219 / 1.064 / 1.801	0	0/36	160.5
Analysis	1.105 [.726, 1.504]	.114 / 0 / .134	−.152	0/36	8466.8
Four-fold CV	1.217 [.754, 1.796]	.002 / 0 / 0	0	0/36	4821.7

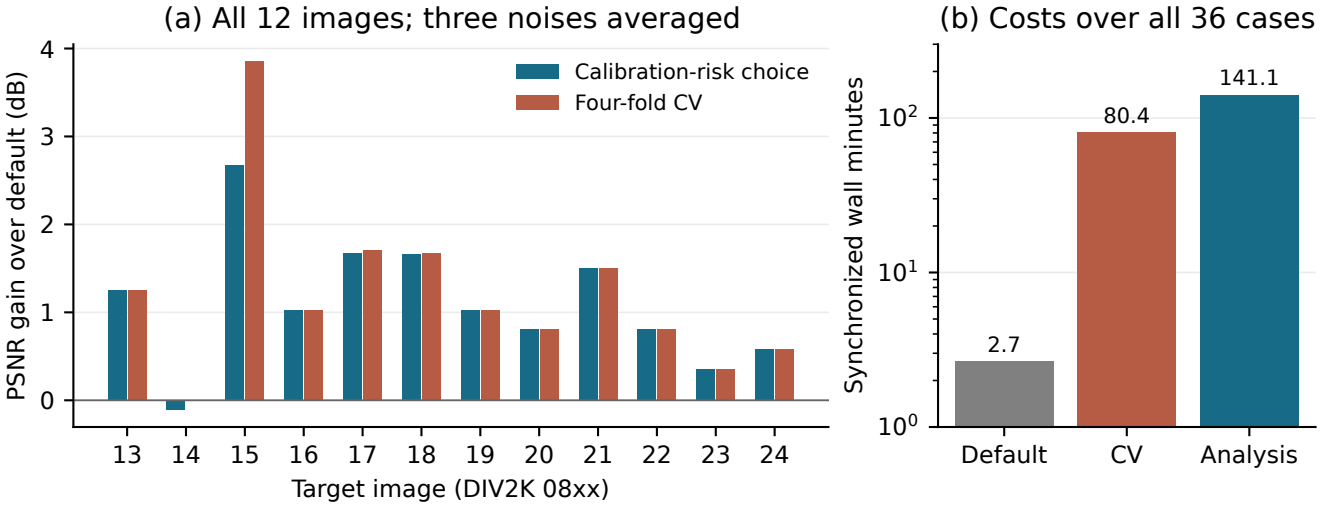


Figure 2: G2 complete-cohort gains and costs. (a) Every target image is shown; averaging three noise settings does not create three independent image replicates. (b) Synchronized wall time of common preparation, selection and final fits on the same A800, on a log axis; this is not GPU busy time. The analysis rule chooses power exponent 2.5 in all 36 cases, so its gain does not establish instance-specific adaptation.

D11's revised bootstrap covers fixed images/noise draws and does not measure training-seed variability.

- **Guidance generalisation.** G2 covers 12 fixed images, three noise settings with fixed draws, one operator and one 32-image calibration. Zero observed severe degradations does not certify a population rate below 5%. Analysis chooses the same prior in all cases, so its improvement over the default does not establish target-specific adaptation. It trails CV in accuracy and cost. Four-probe variability, calibration mismatch and unobserved spectral components remain limitations; no INR architecture selection is established. Oracle candidates were not all saved as pixels.
- **Action-class dependence.** Unidentifiable spectra can still permit useful mixture reconstruction. Appendix E distinguishes selecting a fixed candidate from arbitrary estimators; its elementary two-point separation is classical illustrative material, not a general sharp sampling theorem or a nonlinear-network guarantee.
- **EEG evidence.** EEG currently contributes 200 calibration segments; the 760-segment test result has not been recovered and is excluded from confirmatory claims in this revision.

7 Conclusion

Observation-conditioned linear references clarify spectral mismatch costs and their asymmetry. Exact scalar and matrix identities provide the foundation; band approximations and neural interpretations require separate evidence. Completed experiments support mismatch-cost ranking and repeated readout-loss ordering while preserving failures of the earlier winner diagnostic, function equivalence and a proposed gain mechanism. Practical guidance must specify the reference spectrum and operator and demonstrate selection regret against competing baselines. A completed prospective test shows that calibrated analysis can improve a fixed GP default, but its constant choice, higher regret and higher cost than CV leave adaptive and computational advantages unsupported. The formulas are useful conditional diagnostics; deployment benefits require direct comparative evidence.

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A Proof of Proposition 3

With $R = p/(p+q)$ the matched gain R attains error $p(1-R)$. The gain $g = \kappa p/(\kappa p + q) = \kappa R/(\kappa R + 1 - R)$ gives error $(1-g)^2 p + g^2 q$. Using $q/p = (1-R)/R$ and $D = \kappa R + 1 - R$,

$$\frac{(1-g)^2 p + g^2 q}{p} = \frac{(1-R)\big[(1-R) + \kappa^2 R\big]}{D^2}.$$

Subtracting $(1-R)$ and expanding D^2, the numerator becomes $(1-R)\,R(1-R)(\kappa-1)^2$, which is Eq. (2). For Cor. 1, $\Delta(R,\kappa)/\Delta(R,1/\kappa) = \big((1-R)\kappa + R\big)^2/\big(1 - R + \kappa R\big)^2$, which exceeds 1 for $\kappa > 1$ iff $(\kappa-1)(1-2R) > 0$, i.e. iff $R < 1/2$ (also checked numerically).

B Reproducibility

Original sources, protocols and results retain their server directory names in the collected evidence. The updated `SOURCES.md` maps each claim to original or new output. New G1 and L2 runs invoke original frozen evaluation code and preserve old results. The G1 protocol matches SHA256 prefix `548b26fe`; the full hash is recorded in `SOURCES.md`. Independent L2 checking reconstructs pooled recoverability from saved per-image signal/error arrays; G1 checking validates saved pooled numbers and coverage, but per-image energy weights were not saved by its original script. Exact machine-readable numerical checks accompany the evidence.

C Matrix optimality and aggregation counterexamples

Let $e_P = x - H_P y$. Since $\mathbb{E}[e_P y^\top] = 0$, expanding $x - Hy = e_P + (H_P - H)y$ cancels both cross terms. Multiplication by an arbitrary fixed W and taking traces gives the identity in §3. Gaussianity is unnecessary for this second-moment calculation.

Even equal-energy modes with identical $\kappa = 4$ demonstrate that band plug-in costs have no uniform error direction. For $R = (.1, .9)$ the average exact cost is .218639; for $R = (.01, .99)$ it is .041601. Both have average $R = .5$, whose plug-in cost is .18: an underestimate in the first example and an overestimate in the second. These examples do not use non-diagonal sampling, so aggregation alone suffices. For fixed finite $\kappa > 0$ the cost vanishes as $R \to 0$ or 1, but the low-R limit is not uniform in κ: $R = .01$, $\kappa = 100$ gives $\Delta = 24.256863$. The scalar identity was independently checked on a 101×121 grid with maximum algebraic residual 7.2×10^{-15} [src: CPU-10000].

D Exploratory same-observation linear controls

This development diagnostic separates a model's own uncertainty from actual reconstruction error. It uses two already-seen images, DIV2K 0801/0802, one Fourier-feature seed and one paired noise draw; it is neither a confirmatory comparison nor a new selection method. A central native 128×128 crop is average-pooled to 64×64. Two point-sampling frames use vertical stride four and horizontal phases $(0,0)$, $(0,1)$ or $(0,2)$, each with 512 noisy RGB observations and noise variance $\nu = 10^{-4}$. Only horizontal phases change; this matched point operator differs from E1's native integrated footprints.

All 24 linear predictions are fixed before truth-based scoring: two images, three geometries, two scales $\sigma \in \{3, 10\}$ and two kernels. The RBF is $k_\sigma(u,v) = \exp[-2\pi^2\sigma^2\|u-v\|^2]$. The finite Fourier kernel uses the original network's 256 frozen Gaussian frequencies, with $\phi(u) = [\cos(2\pi Bu), \sin(2\pi Bu)]/\sqrt{256}$ and $k(u,v) = \phi(u)^\top \phi(v)$. Both have unit diagonal, fixed mean $m = .5$ and ridge parameter ν, solved in float64 by Cholesky. No bandwidth, amplitude or regularisation search is performed. Saved nonlinear FF outputs are compared without retraining. Fixed Fourier ridge is not the MLP's full NTK or learned kernel [Tancik et al., 2020]; its mean, regularisation and optimisation also differ from those of the FF network. The noise generator resets to the same seed for each geometry, giving common random numbers across geometries.

Let $T = K_{to}(K_{oo} + \nu I)^{-1}$ for the P scored target pixels. Conditioning on a fixed clean RGB image, the exact raw-output risk over fresh observation noise is

$$\frac{\|T(X_o - m) - (X_t - m)\|_F^2}{3P} + \frac{\nu\|T\|_F^2}{P}.$$

Truth is used only to score the bias term after the complete prediction cohort is hash-frozen. For the saved noise draw, actual raw MSE further includes its realised noise term and signed bias–noise cross term. These identities do not apply after clamping, and are not observation-only risk estimators. In contrast, $\mathrm{diag}(K_{tt} - TK_{ot})$ is

Estimator	σ	0801	0802
RBF	3	+.763	+.552
RBF	10	+1.792	+1.908
Fixed Fourier ridge	3	+.480	−1.125
Fixed Fourier ridge	10	−.647	−.664
Trained FF	3	+.633	+.443
Trained FF	10	+.951	+.919

Table 12: Development difference in clamped PSNR (dB), phase $(0, 2)$ minus $(0, 1)$, on the identical shave-four region. All opposing signs are retained; these two images and one seed do not establish a population ordering or a nonlinear mechanism.

uncertainty under the candidate's own Gaussian prior. Different kernels specify different target laws, so their own uncertainties are not a common actual-risk comparison.

Table 12 has three opposing finite-Fourier comparisons. At $\sigma = 10$, phase $(0, 2)$ has conditional expected *raw* MSE 1.901/2.473 on 0801/0802, predominantly bias; clipping must not conceal this failure. Within each fixed kernel/scale, model posterior variance orders the three phases as $02 < 01 < 00$, while the conditional noise propagation term orders them as $00 < 02 < 01$. Total conditional raw-risk ordering agrees with model variance in only five of eight image/kernel/ scale combinations. Lower self-prior uncertainty therefore does not certify lower fixed-image error.

On the discrete 56×56 target crop, phases $(0, 1)$ and $(0, 2)$ have exactly the same nearest-observation distance histogram. Continuous periodic maximum gaps three/two alone cannot explain their different errors, especially for these nonperiodic kernels and nonnested observation sets. Covariance-parameter identifiability and reconstruction error are different objects; this diagnostic supplies no general transfer theorem. Joint correlations, prior mismatch and estimator behaviour require separate analysis rather than a nearest-distance or rank-only mechanism.

All 24 controls finish with no numerical failure in 17.39 seconds of shared CPU wall time (four threads, nice 10, 600-second absolute budget). The complete prediction manifest precedes scoring; all saved outputs, linear responses, residuals and failures are retained. Independent LU recomputation of all 24 controls and primal/dual checking of all 12 Fourier controls agree with predictions within 2.51×10^{-11}. These audits validate the calculations, not natural-image prior validity, practical guidance or publication novelty [src: CTRL-20261008].

As a common-law diagnostic, the same 12 distinct saved linear responses are evaluated on 32 separate calibration crops, DIV2K 0001–0032, with identical preprocessing, mean, noise variance and target region. Averaging the preceding conditional-risk formula over these complete RGB images defines a finite empirical evaluation law; no Gaussian or stationarity assumption or population confidence interval is attached to it. All 384 image/operator risks and eight static recommendations are hash-frozen before consulting the two development images. This is a retrospective diagnostic designed after their earlier outcomes, not prospective evidence. Calibration chooses RBF $\sigma = 3$ within every geometry, and phase $(0, 2)$ globally, agreeing with the conditional-risk optimum on both development images. Across the overlapping eight choice groups per image, 15 of 16 recommendations match their oracle; the exception is Fourier ridge $\sigma = 3$ on 0801, with raw-risk regret .001037. Nevertheless, 15 of 132 pairwise comparisons have opposing signs. These dependent comparisons do not establish a general transfer rate. The calibration recommendation within each geometry coincides with the fixed RBF $\sigma = 3$ baseline; there is no image-adaptive gain. The complete CPU diagnostic takes 8.23 seconds without solving new systems or retraining [src: CAL-20261008].

E Ambiguity, selection and unrestricted reconstruction

This elementary control illustrates classical robust decision principles, not a new minimax theorem [Kitagawa, 2011]. On eight periodic sites let $Q_k(i, j) = \frac{1}{4}\cos[2\pi k(i - j)/8]$, $D = Q_1 - Q_3$, and $C_\pm = I \pm \theta D$, with known $0 < \theta < 1$, unknown sign, zero mean and known noise variance $\nu > 0$. The priors are positive definite. Write $a = 1 + \nu$ and let A_{01} and A_{02} select sites $(0, 4, 1, 5)$ and $(0, 4, 2, 6)$ respectively. All risks below are total eight-coordinate squared error; divide by eight for per-coordinate MSE.

For $A = A_{02}$, $ADA^\top = 0$ and $\mathrm{tr}(AD^2A^\top) = 2$. Both worlds therefore have exactly the same observed Gaussian law, $N(0, aI)$, and posterior means $m_\pm = A^\top y/a \pm \theta DA^\top y/a$. Regret subtracts each world's own known-sign Bayes risk. Their midpoint is an admissible mixture estimate. Posterior orthogonality and the parallelogram identity give

$$\frac{\mathrm{Reg}_+(g) + \mathrm{Reg}_-(g)}{2} = \mathbb{E}\|g - A^\top y/a\|^2 + \frac{2\theta^2}{a}.$$

The midpoint attains both regrets equally, hence the unrestricted minimax regret is $2\theta^2/a$ and total minimax risk is $8 - 4/a$. Choosing only between m_+ and m_- instead gives minimax regret $4\theta^2/a$ when randomisation is allowed; fixed wrong selection costs $8\theta^2/a$. Candidate-selection failure must not be promoted to the same lower bound for all reconstruction methods.

The known-spectrum oracle favours A_{02} by the positive risk gap

$$\delta = R^*_{01} - R^*_{02} = \frac{\theta^2(1 + 2\nu - \theta^2)}{a(a^2 - \theta^2/2)} < \frac{\theta^2}{a}.$$

For K independent calibration draws observed through the chosen geometry, followed by a fresh evaluation draw,

both designs pay $4(K+1)$ measurements. Geometry 02 cannot learn the sign for any K. Geometry 01 has distinct laws; calibration-only likelihood selection has error at most $\frac{1}{2}\exp[-K\theta^2/(4a^2)]$ and wrong-rule cost $\Gamma = 8a\theta^2(a^2 - a + 1/2)/(a^2 - \theta^2/2)^2$. It beats geometry 02's unrestricted worst-case total risk whenever $\Gamma\exp[-K\theta^2/(4a^2)]/2 < 2\theta^2/a - \delta$. This is a sufficient finite threshold, not a sharp transition or a new general sampling law. Free full-resolution calibration or spectrum labels would invalidate the information restriction. Prediction versus covariance-learning design is already studied [Zimmerman, 2006]; the present two-point illustration supplies no natural-image or INR guarantee [src: SEP-20261008].